BEN BINH TRAN

RF & Microwave Engineer | Electrical & Computer Engineering

Antenna Design | SATCOM | Phased Arrays | Computational EM | RF Systems

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Summary & Objective

PhD-level Electrical and Computer Engineer with deep expertise in RF/microwave systems, antenna design, satellite communications, and computational electromagnetics. Proven research record spanning broadband dielectric resonator antenna arrays, radome material characterization, electronically steerable phased arrays, and microwave imaging — with IEEE publications and Raytheon-sponsored project experience. Brings complementary strengths in quality systems, statistical analysis, operations, and workflow automation. Bilingual English/Vietnamese with 20+ years of U.S. engineering education and professional experience. Interactive CV: benbinhtran.github.io/TSE

Professional Experience

Research Assistant (SATCOM) — Brigham Young University / Linear Signal Inc. (Raytheon Sponsored)
Provo, UT, USA | 2010 – 2014

- **Satellite Communication Antenna Array:** Led end-to-end design, HFSS simulation, and fabrication of a broadband, high-efficiency dielectric resonator antenna (DRA) array for SATCOM; achieved a 10% improvement in aperture efficiency over existing models.
- **Radome Material Characterization:** Developed and validated an electrodynamic model for low-loss materials at microwave frequencies and high temperatures, owning the testing process to provide critical performance data for radome design.
- **Active Steerable Antenna Array:** Designed and optimized an electronically steerable array with dual RHCP/LHCP circular-polarization capabilities for enhanced satellite link performance.

Research Assistant — University of Arkansas (MOLICS Lab)
Fayetteville, AR, USA | 2005 – 2010

- **Microwave Imaging for Object Detection:** Applied advanced microwave imaging methods for non-contact object detection and characterized novel materials for tissue-mimicking phantoms used in system validation.
- **RF Device Fabrication & Testing:** Fabricated and tested RF devices and components under strict laboratory safety and operational protocols.

Quality Control & Assurance Specialist — Black Diamond & Kizik
USA | 2022 – 2023

- Conducted statistical analysis of test data to determine product and component strength ratings across typical use, misuse, and abuse scenarios.
- Drove investigations pertaining to proposed design changes, material qualifications, and manufacturing issues; assisted in creating testing models for new products, raw materials, and finished goods.
- Assisted in the fabrication and development of test fixtures for production monitoring while maintaining comprehensive quality record systems for test, design, and product rating documents.

FedEx Vietnam — Customer Experience Agent (RPI Creator & Monitor)
Ho Chi Minh City, Vietnam | 02/2026 – 08/2026

- Created RPI C-MOD control numbers in CEBS and submitted SharePoint entries triggering bilingual consignee notifications; validated 12-digit AWB data and attached RPI, commercial, MSDS, and related documentation.
- Owned the case lifecycle across SharePoint, Excel, ServiceNow, and CEBS; monitored exception queues, flagged SLA risks, and escalated through Station Supervisor, Operations Manager, and Regional RPI paths.
- Implemented Power Automate flows to consolidate AWB rows, normalize multi-recipient fields, group AWBs by recipient, and send one consolidated HTML email; used Power Query and SharePoint logs to track case aging and SLA metrics.
- Coordinated with China and Taiwan teams for RPI monitoring and with Thailand teams for RPI creation, supporting consistent regional handoffs and operational visibility.

FedEx Vietnam — Customer Experience Agent / Supplies Coordination
Ho Chi Minh City, Vietnam | 02/2026 – 05/2026

- Collaborated with Malaysia and Singapore teams to coordinate supplies ordering and delivery; documented and standardized order placement, tracking, and escalation procedures.
- Instituted a “Data Not Found” check for missing or delayed orders, escalated unresolved account issues to Operations, and authored bilingual booking-confirmation and order-status templates.

Customer Service Partner Specialist — Concentrix
Ho Chi Minh City, Vietnam | 2024 – 2025

- Provided B2B platform support, technical triage, billing coordination, and high-priority dispute resolution; escalated defects to engineering and logged root causes in ServiceNow.

Shift Lead / Sales Manager — AutoZone Inc.
USA | 2018 – 2020

- Owned assigned-shift operations; supervised and trained a team of 6+ across merchandising, sales, safety, cash handling, and inventory control; managed Commercial/Hub fulfillment for corporate accounts.

RF & Engineering Research Highlights

Antenna Array Design

Broadband DRA array for SATCOM with 10% aperture efficiency gain over prior models; HFSS simulation and full fabrication.

Radome Characterization

Electrodynamic model for low-loss materials at microwave frequencies and high temperatures; Raytheon-sponsored testing process ownership.

Steerable Phased Array

Electronically steerable array with dual RHCP/LHCP circular polarization for enhanced satellite link performance.

Microwave Imaging

Non-contact object detection via microwave imaging; material characterization for tissue-mimicking phantoms in system validation.

Core Competencies

- **RF & Satellite Systems:** Phased array antennas, DRA design, radome analysis, microwave imaging, material characterization, link budget, monopulse radar.
- **Engineering Software:** Ansys HFSS, Keysight ADS, CST Empire, MATLAB, LabVIEW, CAD, FEA.
- **Quality & Manufacturing:** QMS, SPC, DFM, root-cause analysis, process validation, failure analysis, test fixture development.
- **Operations & Analytics:** SLA/KPI tracking, Power Query, Excel, Power Automate, ServiceNow, SharePoint, CEBS.

Skills & Tools

Ansys HFSS | Keysight ADS | CST Empire | MATLAB | LabVIEW | CAD / FEA | QMS / SPC | Root-Cause Analysis | Excel / Power Query | Power Automate | ServiceNow | SharePoint | CEBS | SQL / PostgreSQL | JavaScript / HTML

Education

PhD studies (incomplete), Electrical & Computer Engineering
Brigham Young University — Provo, UT | 2010–2014
Radar & Satellite Communications

MS, Electrical Engineering
University of Arkansas — Fayetteville, AR | 2008–2010
Microwave Imaging for Object Detection

BS, Electrical Engineering & Mathematics
University of Arkansas — Fayetteville, AR | 2004–2008

Honors & Awards

- PhD Fellowship — Brigham Young University
- MTT-S IEEE Scholarship
- Travel Fellowship — URSI
- ICT Outstanding Award
- Ambassador of Good Will — AR Governor
- Dean's List & Chancellor's List — U of A
- Honor Society: Gamma Beta Phi, HKN, NSCS
- FIES, MLK & Poster Stone Coop Scholarships

Languages

English (professional — 20+ years U.S.) | Vietnamese (native)

Selected Engineering & Automation Projects

Unified Logistics Workflow — Bulk Booking Suite
Jun 2026 | Vanilla JS, HTML/CSS, Regex parsing, Unit conversion

- Browser-based tools for booking generation, driver notes, and bulk shipment requests with CED/eOPS parsing, weight/dimension checks, and Outlook-ready approval-email output.
- Demo: *bulkbooking.html* | README: *BT-utility-tools-demo*

Order Automation System — Logistics Supplies Management
Jun 2026 | Node/Express, HTML/CSS/JS

- Parses structured text, enforces category/item limits, prevents duplicate pending orders, and generates audit-ready order histories.
- Demo: *orders.html* | README: *BT-utility-tools-demo*

Customer Experience Request Management Automation Platform
Jun 2026 | Power Automate, Power Apps, Microsoft Lists, Outlook, M365

- Case-management and workflow-automation platform connecting intake, routing, SLA tracking, and bilingual stakeholder notifications; AWB Consolidation Flow groups recipients and sends one consolidated HTML email.
- README: *BT-utility-tools-demo*

Case Library & Templates Organizer
Jul 2026 | Vanilla JS, HTML/CSS, localStorage

- Searchable repository of bilingual customer-service scenarios, email templates, and internal notes with live editing and locally saved customizations.
- Demo: *caselibrary.html* | README: *BT-utility-tools-demo*

Publications

- **"High Efficiency Dielectric Resonator Antenna Array Feeds for Satellite Communications."** *IEEE Transactions on Antennas & Propagation*, 2014.
- **"Broadband, High Efficiency Dielectric Resonator Antenna for Satellite Communications."** *IEEE APS-URSI Symposium*, 2013.